UNIVERSITY OF SOUTH CAROLINA
Electrical Engineering

FINAL EXAMINATION FOR

(Spring Semester: 2022)

ELCT 361– ELECTROMAGNETICS

April 29th, 2022 – Time Allowed: 2 hours (09:00 AM ~ 11:00 AM)

INSTRUCTIONS TO CANDIDATES

1. This paper contains **FIVE (5)** questions and comprises **TWELVE (12)** printed pages.
2. Answer for all problems.
3. Each question carries the same **POINTS**.
4. Please write your answers directly in the **EXAM PAPER**
5. If you make any assumptions in addition to those stated in the question, please state them explicitly and clearly.

< Solutions >

Student ID:

Name:

Q1 **[20 points]** Assume that N turns of wire are wound around a toroidal core of a ferromagnetic material with permeability μ. The core has a mean radius r_0, a circular cross section of radius ($a \ll r_0$), a cross-sectional area S, and a narrow air gap with permeability μ_0 and gap length, l_g, as shown in *Fig. Q1*. A steady current I_0 flows in the wire.

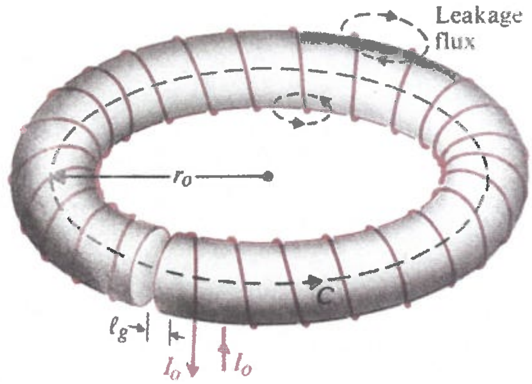


Fig. Q1 Coil on ferromagnetic toroid with air gap

(a) **[8 pts]** Find the magnetic field intensity in the ferromagnetic toroid $\overrightarrow{H_f}$ and in the narrow air gap, $\overrightarrow{H_g}$.

(b) **[12 pts]** Determine the circuit components and draw the magnetic circuit for the toroid shown in *Fig. Q1*.

(a) Based on the Ampere's Law, (N: turns, I_0: current), core (μ) gap (μ_0)

$$\int_S (\nabla \times \vec{H}) \cdot d\vec{s} = \oint_C \vec{H} \cdot d\vec{l} = N I_0$$

$l_g \ll l_c$, $\underline{\vec{B_f} \approx \vec{B_g} = \hat{a}_\phi B_f}$, ($B_f$: magnetic flux density in the ferromagnetic core, B_g: magnetic flux density in the gap). Therefore,

$$\vec{H_f} = \hat{a}_\phi \frac{B_f}{\mu} \quad \text{and} \quad \vec{H_g} = \hat{a}_\phi \frac{B_g}{\mu_0} = \hat{a}_\phi \frac{B_f}{\mu_0}$$

$$\oint_C \vec{H} \cdot d\vec{l} = \oint (\hat{a}_\phi \vec{H_f} + \hat{a}_\phi \vec{H_g}) \cdot d\vec{l} = \frac{B_f}{\mu}(2\pi r_0 - l_g) + \frac{B_f}{\mu_0} l_g = N I_0$$

So, $$\vec{B_f} = \hat{a}_\phi \frac{\mu_0 \mu N I_0}{\mu_0 (2\pi r_0 - l_g) + \mu l_g} \quad (\vec{B_f} = \mu \vec{H_f},\ \vec{B_g} = \mu_0 \vec{H_g})$$

Therefore,

$$\vec{H_f} = \frac{\mu_0 N I_0}{\mu_0 (2\pi r_0 - l_g) + \mu l_g}$$

$$\vec{H_g} = \frac{\mu N I_0}{\mu_0 (2\pi r_0 - l_g) + \mu l_g}$$

(b) To analyze the gap toroid based on magnetic circuit.

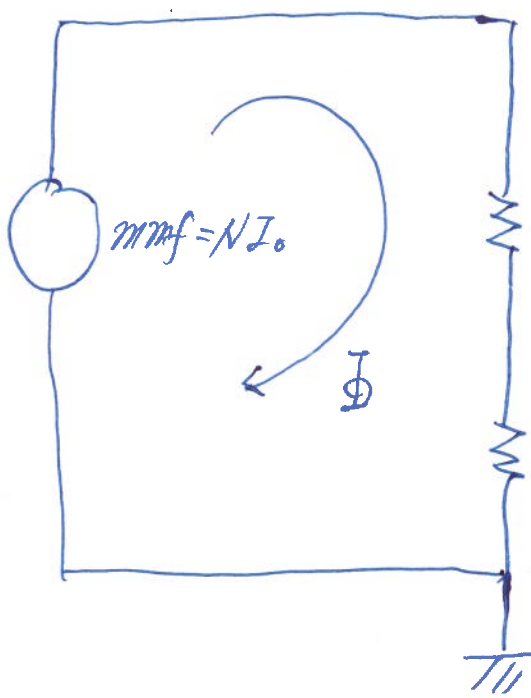
① mmf, ② $\mathcal{R}$ ③ Φ.

(i) $\mathcal{R}$ (for ferromagnetic core) $= \dfrac{2\pi r_0 - l_g}{\mu S}$, $\mathcal{R}$ (for gap) $= \dfrac{l_g}{\mu_0 S}$

(ii) $\Phi = \int_S \vec{B} \cdot d\vec{s} = BS = \dfrac{\mu_0 \mu N I_0}{\mu_0(2\pi r_0 - l_g) + \mu l_g} \cdot S$

$$= \frac{N I_0}{\frac{(2\pi r_0 - l_g)}{\mu S} + \frac{l_g}{\mu_0 S}} = \frac{mmf}{\mathcal{R}_{fc} + \mathcal{R}_g}$$

(iii) $mmf = N I_0$



$\mathcal{R}_{f(core)} = \dfrac{2\pi r_0 - l_g}{\mu S}$ $V_{\mathcal{R}_{f,core}} \Rightarrow$

$\mathcal{R}_g = \dfrac{l_g}{\mu_0 S}$

$$\Phi = \frac{N I_0}{\frac{(2\pi r_0 - l_g)}{\mu S} + \frac{l_g}{\mu_0 S}}$$

Q2 (a) **[8 pts]** Find the induced voltage in the conductor of *Fig. Q2-(i)* where $\vec{B} = \mathbf{0.04}\,\widehat{a_y}$ **[T]** and the velocity of conducting wire is $\vec{U} = \mathbf{2.5sin10^3t\widehat{a_z}}$ **[m/s]**.

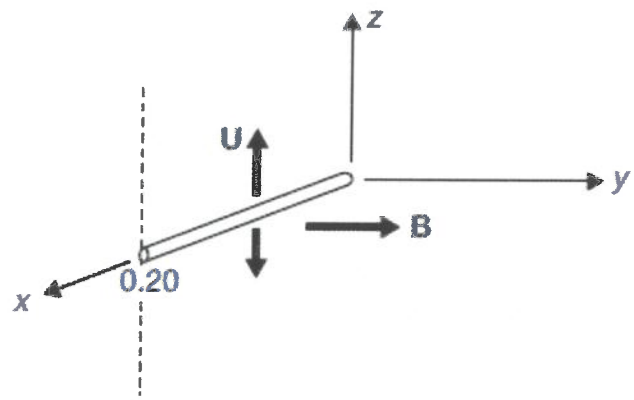


Fig. Q2-(i)

(b) **[12 pts]** The rectangular coil in *Fig. Q2-(ii)* is in a field

$$\vec{B} = 0.05\frac{\widehat{a_x} + \widehat{a_y}}{\sqrt{2}}\ [T]$$

Find the torque about the z axis when the coil is in the position shown and carries a current of **5 A.**

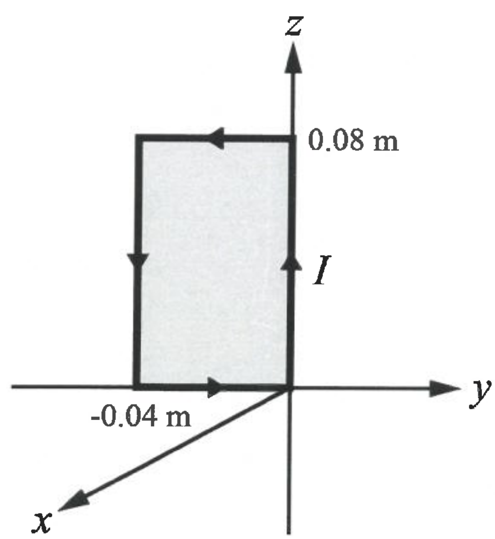


Fig. Q2-(ii)

(a) The induced voltage → $\int_{initial}^{final} (\vec{u} \times \vec{B}) \cdot d\ell$

where, $\vec{u} = 2.5 \sin 10^3 t\, \hat{a}_z$ (m/s), $\vec{B} = 0.04 \hat{a}_y$ [T]

Therefore,

$$v(\text{Voltage}) = \int_0^{0.20} (2.5 \sin 10^3 t\, \hat{a}_z) \times (0.04\, \hat{a}_y) \cdot \hat{a}_x\, dx$$

$$= \int_0^{0.20} 0.10 \sin 10^3 t\, (-\hat{a}_x) \cdot \hat{a}_x\, dx = -0.02 \sin 10^3 t\ (V)$$

(The conductor first move in the $\hat{a}_z$ direction.) The $x = 0.20$ end is negative with respect to the end at the z axis for this half circle.

(b) $\vec{m}$ (lever arm) $= IA\, \hat{a}_n = 1.6 \times 10^{-2} \hat{a}_x$

$$\vec{T} = \vec{m} \times \vec{B} = 1.6 \times 10^{-2} (\hat{a}_x) \times 0.05 \frac{\hat{a}_x + \hat{a}_z}{\sqrt{2}}$$

$$= 5.66 \times 10^{-4} \hat{a}_z\ [N/m]$$

Q3 **[20 pts]** A coaxial capacitor shown in *Fig. Q3* with inner radius $\boldsymbol{a}$ = **5 [mm]**, outer radius $\boldsymbol{b}$ = **6 [mm]**, and length $\boldsymbol{L}$ = **500 [mm]** has a dielectric for $\boldsymbol{\epsilon_r = 6.7}$, a permeability of $\boldsymbol{\mu_r = 0.5}$, and a conductivity of **45 [mS/m]**. A $\boldsymbol{v(t) = 250 \sin 377t}$ **[V]** is applied for this capacitor. [The capacitor has $2\pi\epsilon L/\ln{(b/a)}$ of capacitance]

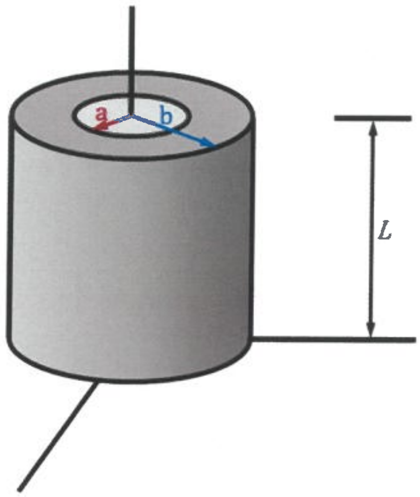


Fig. Q3 A coaxial capacitor

(a) **[10 pts]** Find the displacement current, $\boldsymbol{i_D}$, in the capacitor due to the applied sinusoidal voltage source and compare with conduction current, $\boldsymbol{i_c}$

(b) **[5 pts]** If the magnitude of conduction current density $(\overrightarrow{J_c})$ is the same as the $\overrightarrow{J_D}$, what is the frequency of the applied sinusoidal voltage source?

(c) **[5 pts]** Find the inductance between the inner and outer conductors.

$v(t) = 250 \sin 377t$ [V] to cylindrical capacitor. To determine the potential between the inner and the outer conductors, Laplace's Equation is applied. $\nabla^2 V = 0$ (potential is as a function of r)

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V}{\partial r}\right) = 0 \rightarrow r\frac{\partial V}{\partial r} = A, \quad \text{So } V = A\ln[r] + B$$

two Boundary conditions, $V = 0$ $(r = 0.005)$ $V = 250 \sin 377t$ $(r = 0.006)$

$$A = \frac{250 \sin 377t}{\ln\left(\frac{6}{5}\right)} \qquad B = -\frac{250 \sin 377t}{\ln\left(\frac{6}{5}\right)} \ln(0.005)$$

Therefore, $$V = \frac{250 \sin 377t}{\ln\left(\frac{6}{5}\right)} \ln\left[\frac{r}{0.005}\right] \text{ [V]}$$

(a) $i_D = J_D S$ $(S = 2\pi r L)$, $\vec{J_D} = \frac{\partial \vec{D}}{\partial t} = \epsilon_0 \epsilon_r \frac{\partial \vec{E}}{\partial t}$

$$\vec{E} = -\nabla V = -\frac{1.37 \times 10^3}{r} \sin 377t \, \hat{a}_r \text{ [V/m]}$$

Therefore, $\vec{J}_D = \epsilon_0\epsilon_r \frac{\partial \vec{E}}{\partial t} = \left(\frac{1}{36\pi}\times10^{-9}\right)(6.7)\frac{\partial}{\partial t}\left(\frac{1.37\times10^{3}}{r}\sin 377t\,\hat{a}_r\right)$

$= -\frac{3.07\times10^{-5}}{r}\cos 377t\,\hat{a}_r\ (A/m^2)$

$\therefore i_D = J_D(2\pi r L) = +\frac{3.07\times10^{-5}}{r}\cos 377t\times(2\pi)\times(0.001)\times(0.5)$

$= 9.63\times10^{-5}\cos 377t\ (A)$

$C = \frac{2\pi\epsilon_0\epsilon_r L}{\ln\left(\frac{b}{a}\right)} = \frac{2\pi\epsilon_0(6.7)(0.5)}{\ln\left(\frac{6}{5}\right)} = 1.02nF = 1.02\times10^{-9}F$

Then $i_C = C\frac{dV}{dt} = (1.02\times10^{-9})(250\times377)(\cos 377t) = 9.63\times10^{-5}\cos 377t\,(A)$

$i_C \approx i_D$

(b) $|\vec{J}_C| = |\vec{J}_D| \rightarrow \sigma = \omega\epsilon \rightarrow f = \frac{43\times10^{-3}}{2\pi\epsilon_0(6.7)} \doteq 1.22\times10^{8}\ Hz$ or $0.122\ GHz$

(c) $L = \frac{\Lambda(I)}{I}$, $\vec{B} = \frac{\mu_0\mu_r I}{2\pi r}\hat{a}_\phi$

$\Phi = \int_S \vec{B}\cdot d\vec{s} = \int_0^L\int_a^b \frac{\mu_0\mu_r I}{2\pi r}\,dr\,dz = \frac{\mu_0\mu_r I L}{2\pi}\ln\left(\frac{b}{a}\right)$

$= \int_0^{0.5}\int_{0.005}^{0.006}\frac{\mu_0\mu_r I L}{2\pi r}\,dr\,dz.$

Therefore

$L = \frac{\Phi}{I} = \frac{\mu_0\mu_r L}{2\pi}\ln\frac{b}{a} = \frac{(4\pi\times10^{-7})(0.5)(0.5)}{2\pi}\ln\left(\frac{6}{5}\right)$

$= 9\times10^{-9}\,[H/m]$

Q4 **[20 pts]** A time-harmonic electric field, $\vec{E}(t) = E_m \sin(\omega t - \beta z)\,\widehat{a_y}$ is traveling in the air along the z −direction and is absorbed by a simple medium with $\epsilon_r = 20$ and $\mu_r = 5$. Where, $\mu_0 = 4\pi \times 10^{-7}\ [H/m]$ and $\epsilon_0 = \frac{1}{36\pi} \times 10^{-9}[F/m]$.

(a) **[3 pts]** Find displacement current density, $\overrightarrow{J_D}$ in the medium

(b) **[4 pts]** If a time-harmonic magnetic field, $\vec{H}(t)$, due the $\vec{E}(t)$ in the medium is generated as $H_m sin\ (\omega t - \beta z)\widehat{a_x}$, Find H_m.

(c) **[6 pts]** Find the wave velocity, u and wave number k in the medium if the absorbed $\vec{E}(t)$ has a frequency of *1 GHz*.

(d) **[7 pts]** How much percent of $\vec{E}(t) = E_m \sin(\omega t - \beta z)\,\widehat{a_y}$ in the air is transmitted in the medium?

(a) In the medium,

$$\vec{J}_D = \epsilon \frac{\partial \vec{E}}{\partial t} = \epsilon_0 \epsilon_r \frac{\partial}{\partial t}[E_m \sin(\omega t - \beta z)]\,\hat{a}_y$$

$$= \frac{10^{-8}}{18\pi} E_m \omega \cos(\omega t - \beta z)\,\hat{a}_y\ [A/m^2]$$

(b) Using Maxwell's Equation. $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

$$\begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & E_m \sin(\omega t - \beta z) & 0 \end{vmatrix} = -\frac{\partial \vec{B}}{\partial t}$$

$\vec{B}(t) = -\frac{\beta E_m}{\omega} \sin(\omega t - \beta z)\,\hat{a}_x$, hence $\vec{H} = \frac{\vec{B}(t)}{\mu_0 \mu_r}$ in the medium

So, $\vec{H}(t) = -\frac{\beta E_m}{\mu_0 \mu_r \omega} \sin(\omega t - \beta z)\,\hat{a}_x$ $(\mu_r = 5)$

Therefore, $H_m = -\frac{\beta E_m}{5 \mu_0 \omega}$

(c) In the medium,

$$u = \sqrt{\frac{1}{\mu\epsilon}} = \sqrt{\frac{1}{\mu_0\mu_r\epsilon_0\epsilon_r}} = \sqrt{\frac{1}{\mu_0\epsilon_0}}\sqrt{\frac{1}{\mu_r\epsilon_r}}$$

$$= 3\times10^8(m/sec)\cdot\sqrt{\frac{1}{20\cdot5}} = 3\times10^7[m/sec]$$

(d) For transmission, characteristics Impedance of EM Wave in the air must be matched with the characteristics Impedance of medium.

$$\tau(\text{transmission}) = 1 - \Gamma(\text{reflectance})$$

Where, $$\Gamma = \frac{\eta_{EM(air)} - \eta_{(Medium)}}{\eta_{EM(air)} + \eta_{(Medium)}}$$

$$\eta_{EM(air)} = \sqrt{\frac{\mu_0}{\epsilon_0}} = 120\pi, \quad \eta_{medium} = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_r\mu_0}{\epsilon_r\epsilon_0}} = 120\pi\cdot\sqrt{\frac{5}{20}}$$

$$= 60\pi$$

$$\Gamma = \frac{120\pi - 60\pi}{120\pi + 60\pi} = \frac{1}{3}$$

Therefore $$\tau = 1 - \frac{1}{3} = \frac{2}{3}\ (66.7\%)$$

So, 66.7% of EM wave will be transmitted to the medium

Q5 **[20 pts]** Brian wanted to prepare his lunch using an unknow frozen food he found from a refrigerator. He put the frozen food in a microwave for defrosting. When he turned on the power of microwave, a time-harmonic electric wave $\vec{E}(t) = 5\cos(\omega t + \frac{\pi}{4})$ with an angular frequency of ***10 MHz*** was applied to the frozen food with $\epsilon = 1.2\epsilon_0$, and loss tangent of ***0.35***, where $\epsilon_0 = \frac{1}{36\pi} \times 10^{-9}[\frac{F}{m}]$.

(a) **[5 pts]** Find the loss (critical) angle of the frozen caused by the time-varying electrical wave of $\vec{E}(t) = 5\cos(\omega t + \frac{\pi}{4})$.

(b) **[6 pts]** Find the conductivity, σ, and the frequency dependent complex permittivity, ϵ'' of the frozen food.

(c) **[4 pts]** Find the average heating power dissipated in the frozen food per cubic meter due to the applied $\vec{E}(t) = 5\cos(\omega t + \frac{\pi}{4})$.

(d) **[5 pts]** What is the electrical conductance nature of this frozen food? such as "Good conductor" or "Good insulator".

(a) The condition of Q5, $E(t) = 5\cos(\omega t + \frac{\pi}{4})$, $f = 10^7 [Hz]$,

$\epsilon_r = \frac{\epsilon}{\epsilon_0} = 1.2$

$\tan\delta_c = 0.35$, $\rightarrow \delta_c = \tan^{-1}(0.35) \doteq 0.337$

(b) $\sigma \Rightarrow \tan\delta_c = \frac{\sigma}{\omega\epsilon} \rightarrow \sigma = \tan\delta_c \, 2\pi f \epsilon_0 \epsilon_r$

$\sigma = 0.35 \times 2\pi \times 10^7 \times \frac{10^{-9}}{36\pi} \times 1.2$

$= 2.333 \times 10^{-4} [S/m]$

(b-2) $\epsilon'' = \frac{\sigma}{\omega} = \frac{\sigma}{2\pi f} = \frac{2.333 \times 10^{-4}}{2\pi \times 10^7} = 0.371 \times 10^{-11} [F/m]$

(c) $p = \frac{1}{2}EJ = \frac{1}{2}\sigma E^2 = \frac{1}{2}(2.333 \times 10^{-4})(5)^2$

$\approx 0.0029 \approx 0.003 [W/m^3]$

(d) σ vs. $\omega\epsilon$

σ of the frozen food, 2.333×10^{-4} [S/m]

$\omega\epsilon$ of the frozen food,

$$\omega\epsilon = 2\pi f \epsilon_0 \epsilon_r = 6.667 \times 10^{-4}$$

Hence,

$\omega\epsilon$ is slightly greater than σ

Therefore,

the frozen food can be considered as

"Just Insulator"

THE END